# UMC 0.11 um Logic and Mixed-Mode Al Advanced Enhancement Standard Performance 1.2V MOSFET Monte Carlo SPICE Model Document

**Version 0.5 Phase1**
**2010/12/27**

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# 1 Introduction

## 1.1 Revision History

**Table 1-1 Revision history**

| Version | Date | Owner | Comment |
|---|---|---|---|
| 0.2_p1 | 2010/07/28 | Shih-Hsin Yeh | This Monte Carlo model includes both process and mis-matching characteristics extracted from model cards as below:<br>L110AE_SP12_V021.mdl<br>L110AE_SP12_V021.lib |
| 0.3_p1 | 2010/09/24 | Shih-Hsin Yeh | Modify W, L to W*0.9 and L*0.9 in Igate model. |
| 0.4_p1 | 2010/11/01 | Laney Fan | 1..Modify ‘l*0.9’ to ‘l*0.9+12e-9’ for igate formula<br>2. Remove m=1 and m factor in igate formula<br>3.rsh*nrd/m' to 'rsh*nrd'<br>4.'rsh*nrs/m' to 'rsh*nrd'<br>5. Add TWELL Monte Carlo model(copy from SP 1.2V NMOS Monte Carlo model) |
| 0.5_p1 | 2010/12/27 | Shih-Hsin Yeh | This Monte Carlo model includes both process and mis-matching characteristics extracted from model cards as below:<br>L110AE_SP12_V051.mdl<br>L110AE_SP12_V051.lib |

## 1.2 General information

This document serves as a reference to the design works of products that will be manufactured by UMC using the following process.

UMC 0.11 um Logic and Mixed-Mode 1P8M Metal Metal Capacitor Al Advanced Enhancement Process

For more information about this process and technology, please see the electrical design rule listed below.

G-02-LOGIC/MIXED_MODE11-1P8M-MMC/AL/L110AE-EDR

The creation, verification, and usage of the compact model of the following devices are presented in this document.

1.2V n-ch MOSFET
1.2V p-ch MOSFET

The following compact model is chosen for the modeling of these devices.

BSIM3V3.2.4

For detailed descriptions of this compact model equations and parameters, please refer to this material.

BSIM3V3.2.4 MOSFET Model User Manual

In BSIM3V3.2.4, there are two types of models that could be used. One is binning model, which means the whole device dimension range is chopped into several "bins" and different model parameter sets are provided for each bin. The other one is global model, which means one scalable parameter set is provided and it could cover the whole device geometry range. Each of these two approaches has advantages over the other.

UMC provides global models for its many benefits and the model accuracy is guaranteed by a quantitative report of model fitting error against the hardware data, which can be found in the following sections.

## 1.3 Model usage

### 1.3.1 Simulators

The compact model is provided in various formats. It's guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of simulator might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor ELDO version 6.10_1.1

### 1.3.2 The definitions of sub-circuit model parameters

After invoking Monte-Carlo model parameters, the compact transistor models evolve to sub-circuit models. The input parameters for sub-circuit model are the same as compact model, except adding three more input parameters (nf, mf & mis_flag) for sub-circuit model.

The following is the parameter list:

- nf: number of gate finger
- m: number of device
- mf: the same as “m”
- mis_flag: the flag parameter to turn on/off mismatching parameters for individual device. 1 is turn-on and 0 is turn-off. Default is 1.
- w: gate finger width
- l: gate finger length
- as: source area per finger
- ad: drain area per finger
- ps: source periphery per finger
- pd: drain periphery per finger

The following examples show how to use these three additional parameters:

*x1 (d g s b) DeviceName w=5u l=1u nf=5 m='5*3' mf='5*3' mis_flag=1*

The above example shows *3* transistors connected in parallel, each transistor has *5* fingers, and each finger has width *5u* and length *1u*. The mismatching parameters are turned-on for these 3 transistors.

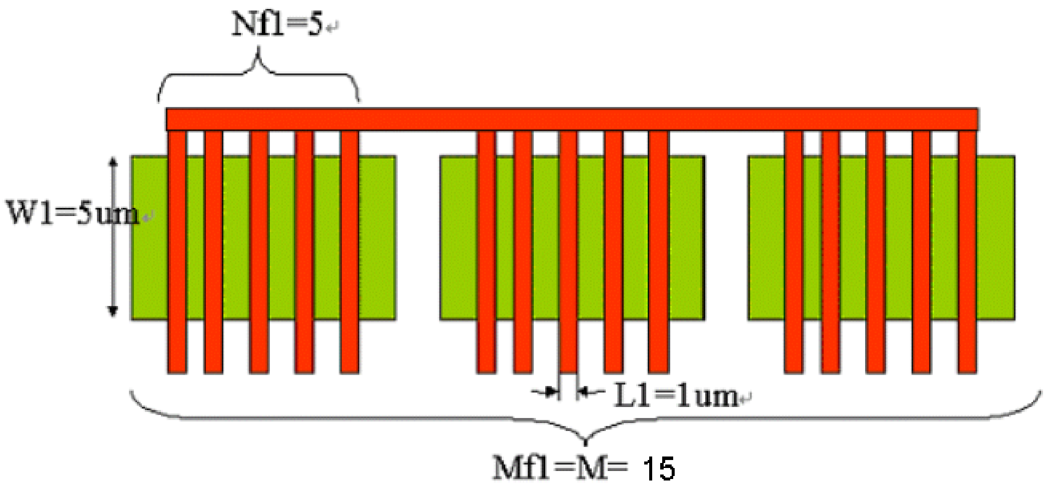


**Figure 1-1 Layout Example of parallel devices.**

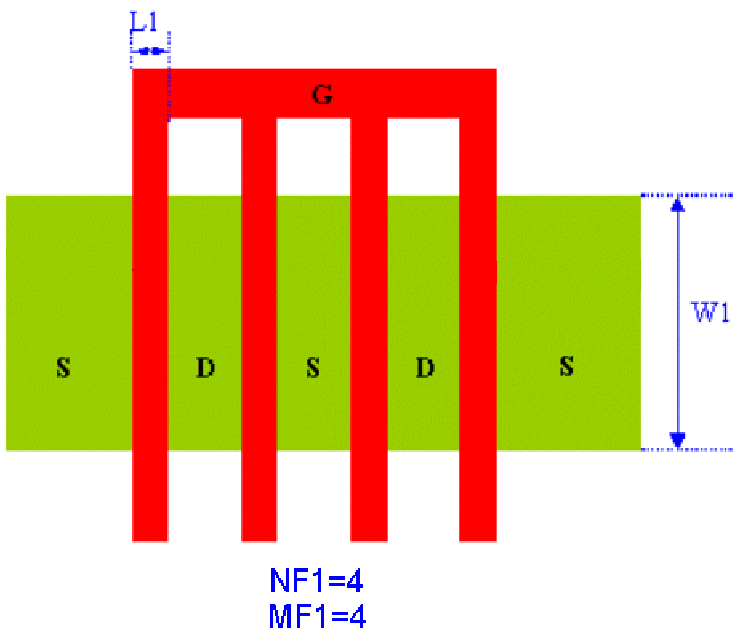


**Figure 1-2 Layout Example of parallel devices.**

### 1.3.3 The combination of statistical model & corner model

The Monte-Carlo model and corner model (5 corners) are combined into a single model library. The user can choose either the Monte-Carlo model or a corner case by selecting different model sections for simulation. The following sections are included:

MC: Monte-Carlo model
TT: Typical n-ch MOSFET
FF: Fast n-ch MOSFET
SS: Slow n-ch MOSFET
FNSP: n-ch MOSFET at middle of Fast and Typical n-ch MOSFET
SNFP: n-ch MOSFET at middle of Slow and Typical n-ch MOSFET
TT_G: Typical n-ch MOSFET (the same as TT)
FF_G: Global corner of Fast n-ch MOSFET
SS_G: Global corner of Slow n-ch MOSFET
SNFP_G: Global corner of n-ch MOSFET at middle of Slow and Typical n-ch MOSFET
FNSP_G: Global corner of n-ch MOSFET at middle of Fast and Typical n-ch MOSFET

### 1.3.4 Mismatching parameters in corner model

The mismatching parameters are included in the corner models. Therefore, the user can run Monte-Carlo mismatching simulation over a corner model. For example, the user can evaluate the mismatching performance at the SS_G case.

### 1.3.5 Sigma number for Monte-Carlo model

The user can define standard deviation at sigma level for Monte-Carlo simulation. If a larger sigma number is used, it means the model parameters will have greater variation. Sigma=3 is recommended to be used.

### 1.3.6 Model usage

**For Synopsys HSPICE**

- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be MC,TT, FF, SS, FNSP, SNFP, FF_G, SS_G, FNSP_G or SNFP_G.

- Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;
Example:
.param sigma=3

- Define device condition by 'Xxx' statement;
Example1: For finger number device (Refer to Figure 1-2):
*X1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1 NF=NF1 M=MF1 MF=MF1*
*+mis_flag=1*
Example2:
*X1 Vdd Vgg Vss Vbb DeviceName W=10e-06 L=1e-06 NF=4 M=4 MF=4*
*+mis_flag=1*

- Define the control of Monte-Carlo simulation.
Indicator:
PROCESS: 1 to turn on the global MC simulation
0 to turn off the global MC process simulation
MISMATCH: 1 to turn on the local MC mismatch simulation
0 to turn off the local MC mismatch simulation

Example 1:
run global & local (total) MC simulation together.
.lib "./ModelFileName.lib" MC
.PARAM PROCESS=1 MISMATCH=1

Example 2:
run global MC simulation only.
.lib "./ModelFileName.lib" MC
.PARAM PROCESS=1 MISMATCH=0

Example 3:
run local MC simulation only.
.lib "./ModelFileName.lib" MC
.PARAM PROCESS=0 MISMATCH=1

Example 4:
run local MC simulation on global corners.
.lib "./ModelFileName.lib" FF_G (or SS_G, FNSP_G,SNFP_G)
.PARAM PROCESS=0 MISMATCH=1

- Define the number of Monte Carlo iterations to run MC simulations.
Example: .DC Vgg 0 1.2 0.01 Sweep Monte=1000

**For Cadence Spectre**

- Add the following statement in the input file to the simulator.
include "./ModelFileName.lib.scs" section=CaseName
, where CaseName could be mc,tt, ff, ss, fnsp, snfp, ff_g, ss_g, fnsp_g or snfp_g.

- Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;
Example:
.param sigma=3

- Define device condition by 'Xxx' statement;
Example1: For Finger Number device ( Refer to Figure 1-2)
*X1 (Vdd Vgg Vss Vbb) DeviceName w=W1 l=L1 nf=NF1 m=mf1 mf=mf1*
*+mis_flag=1*

Example2:
*X1 (Vdd Vgg Vss Vbb) DeviceName w=10e-06 l=1e-06 nf=4 m=4 mf=4*
*+ mis_flag=1*

- Define the number of Monte Carlo iterations and control MC simulations.
Example 1: run global & local (total) MC simulation together.
mc1 montecarlo numruns=1000 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes {
analysisDC1 dc param=Vgg start=0 stop=1.2 step=0.01}
Example 2: run global MC simulation only
mc1 montecarlo numruns=1000 seed=1 variations=process donominal=yes
savefamilyplots=yes {
analysisDC1 dc param=Vgg start=0 stop=1.2 step=0.01}
Example 3: run local simulation only
mc1 montecarlo numruns=1000 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes {
analysisDC1 dc param=Vgg start=0 stop=1.2 step=0.01}
Example 4: run local MC simulation on global corners.
include "./ModelFileName.lib.scs" section= ff_g ( or ss_g, fnsp_g, snfp_g )
mc1 montecarlo numruns=1000 seed=1 variations=mismatch donominal=yes
savefamilyplots=yes {
analysisDC1 dc param=Vgg start=0 stop=1.2 step=0.01}

**For Mentor ELDO**

- Add the following statement in the input file to the simulator.
.lib "./ModelFileName.lib" CaseName
, where CaseName could be MC,TT, FF, SS, FNSP, SNFP, FF_G, SS_G, FNSP_G or SNFP_G.

- Define standard deviation at the sigma level and pick up a value (1~3) of sigma to run MC simulations;
Example:
.param sigma=3

- Define device condition by 'Xxx' statement;
Example1: For finger number device (Refer to Figure 1-2):
*X1 Vdd Vgg Vss Vbb DeviceName W=W1 L=L1 NF=NF1 M=MF1 MF=MF1*
*+mis_flag=1*
Example2:
*X1 Vdd Vgg Vss Vbb DeviceName W=10e-06 L=1e-06 NF=4 M=4 MF=4*
*+mis_flag=1*

- Define the control of Monte-Carlo simulation.
Indicator:
PROCESS: 1 to turn on the global MC simulation
0 to turn off the global MC process simulation
MISMATCH: 1 to turn on the local MC simulation
0 to turn off the local MC simulation
Example 1:
run global & local (total) MC simulation together.
.lib "./ModelFileName.lib.eldo" MC
.PARAM PROCESS=1 MISMATCH=1

Example 2:
run global MC simulation only.
.lib "./ModelFileName.lib.eldo " MC
.PARAM PROCESS=1 MISMATCH=0

Example 3:
run local MC simulation only.
.lib "./ModelFileName.lib.eldo " MC
.PARAM PROCESS=0 MISMATCH=1

Example 4:
run local MC simulation on global corners.
.lib "./ModelFileName.lib.eldo " FF_G (or SS_G, FNSP_G,SNFP_G)

.PARAM PROCESS=0 MISMATCH=1
- Define the number of Monte Carlo iterations to run MC simulations.
Example: .DC Vgg 0 Vcc 1.2 Sweep Monte=1000

Benchmark Model: **l110ae_sp12_v051**
Device Name for 1.2V n-ch MOSFET: **n_12_spl110e_ig**
Device Name for 1.2V p-ch MOSFET: **p_12_spl110e_ig**
Device Name for 1.2V n-ch MOSFET: **n_12bpw_spl110e_ig**

As stated above that one global model is provided and this model is valid within the following geometry, voltage, and temperature ranges. Device behavior beyond these ranges may not be well described by the model and is not guaranteed.

**Geometry range: (Before 90% shrink)**

For 1.2V n-ch MOSFET and p-ch MOSFET:

$0.12\ \mu m \leqq L_{DES} \leqq 50\ \mu m$

$0.16\ \mu m \leqq W_{DES} \leqq 100\ \mu m$

Adopt multi-finger structures if WDES > 9um.
0.306um<=SA,SB<=2.2um ,SD>0.36um
Note:SA,SB=2.2um is the reference dimension of the STI effect.

**Voltage range:**

For 1.2V n-ch MOSFET:

$0V \leqq V_{GS} \leqq 1.2V$ (*1.1)

$0V \leqq V_{DS} \leqq 1.2V$ (*1.1)

$-1.2V \leqq V_{BS} \leqq 0V$

For 1.2V p-ch MOSFET:

$0V \geqq V_{GS} \geqq -1.2V$ (*1.1)

$0V \geqq V_{DS} \geqq -1.2V$ (*1.1)

$1.2V \geqq V_{BS} \geqq 0V$

**Temperature range:**

-50 ℃ ~ +125 ℃

UMC L110AE is to match UMC L110E FEOL device level but have different BEOL Al. option.The following table give users (including designers) a brief guideline of correlation between GDS drawn vs. SPICE/Resistor dimensions.

| Feature\ GDSII | Drawing in shrinkable L130AE GDS database (unit : um) | L110AE GDSII =0.9*(L130AE GDS) Dimension (unit : um) | L110AE SPICE Dimension (unit : um) |
|---|---|---|---|
| Core (W/L) | 10/0.12 | 9/0.108 | HS(L + 8 nm): 9/0.116<br>LL/SP(L +12 nm): 9/0.12 |
| 2.5V I/O (W/L) | 10/0.24 | 9/0.216 | (L +12 nm): 9/0.228 |
| 3.3V I/O (W/L) | 10/0.34 | 9/0.306 | (L +12 nm): 9/0.318 |
| HG 3.3V I/O (W/L) | HG IO-PMOS : 10/0.32<br>HG IO-NMOS : 10/0.34 | HG IO-PMOS : 9/0.228<br>HG IO-NMOS : 9/0.306 | HG IO-PMOS (L +12 nm): 9/0.3<br>HG IO-NMOS (L +12 nm): 9/0.318 |
| Salicide and Non-salicide resistor | (Width/Length) | 0.9*(Width/Length) | 0.9*(Width/Length) |
| Mixed-Mode Device | (Width/Length) | 0.9*(Width/Length) | 0.9*(Width/Length) |
| RF Device | (Width/Length) | 0.9*(Width/Length) | 0.9*(Width/Length) |
| Contact size | 0.16*0.16 | 0.144*0.144 | 0.144*0.144 |
| Via size(1X;2X) | 0.2*0.2; 0.4*0.4 | 0.18*0.18; 0.36*0.36 | MVA1(W -0.02um) : 0.16*0.2<br>Other 1X MVA : 0.18*0.18<br>2X MVA (W -0.04um): 0.28*0.44 |
| Metal size | Width/Space | 0.9*(Width/Space) | 0.9*(Width/Space) |
| Sloting inserting | Metal line width>5.556um | Metal line width>5um | |

**Note:**

1. Because GDS drawn and physical dimension which process dominate are different, please refer to interconnect capacitance model (G-04-LOGIC/MIXED_MODE11-1P8M-MMC/AL-INTERCAP) for precise interconnection-capacitance extraction.
2. Please refer to SPICE model (G-05-LOGIC/MIXED_MODE11-MMC/AL/L110AE-SPICE) for gating precise device characteristics.

# 2 Monte Carlo DC Simulation Result for Process Variation

One thousand Monte Carlo iterations are performed in the following simulation conditions, and the following corner points show TT, SS, FF, SNFP and FNSP values individually.
A reasonable number of Monte Carlo iterations is 30. The statistical significance of 30 iterations is quite high. If the circuit operates correctly for all 30 iterations, there is 99% probability that over 80% of all possible component values operate correctly. The relative error of a quantity determined through Monte Carlo analysis is proportional to (iteration times)^(-1/2).

## 2.1 Threshold voltage in linear region VTLIN

The following figures show the Monte Carlo simulation result of VTLIN threshold voltage extracted at 25 ℃.

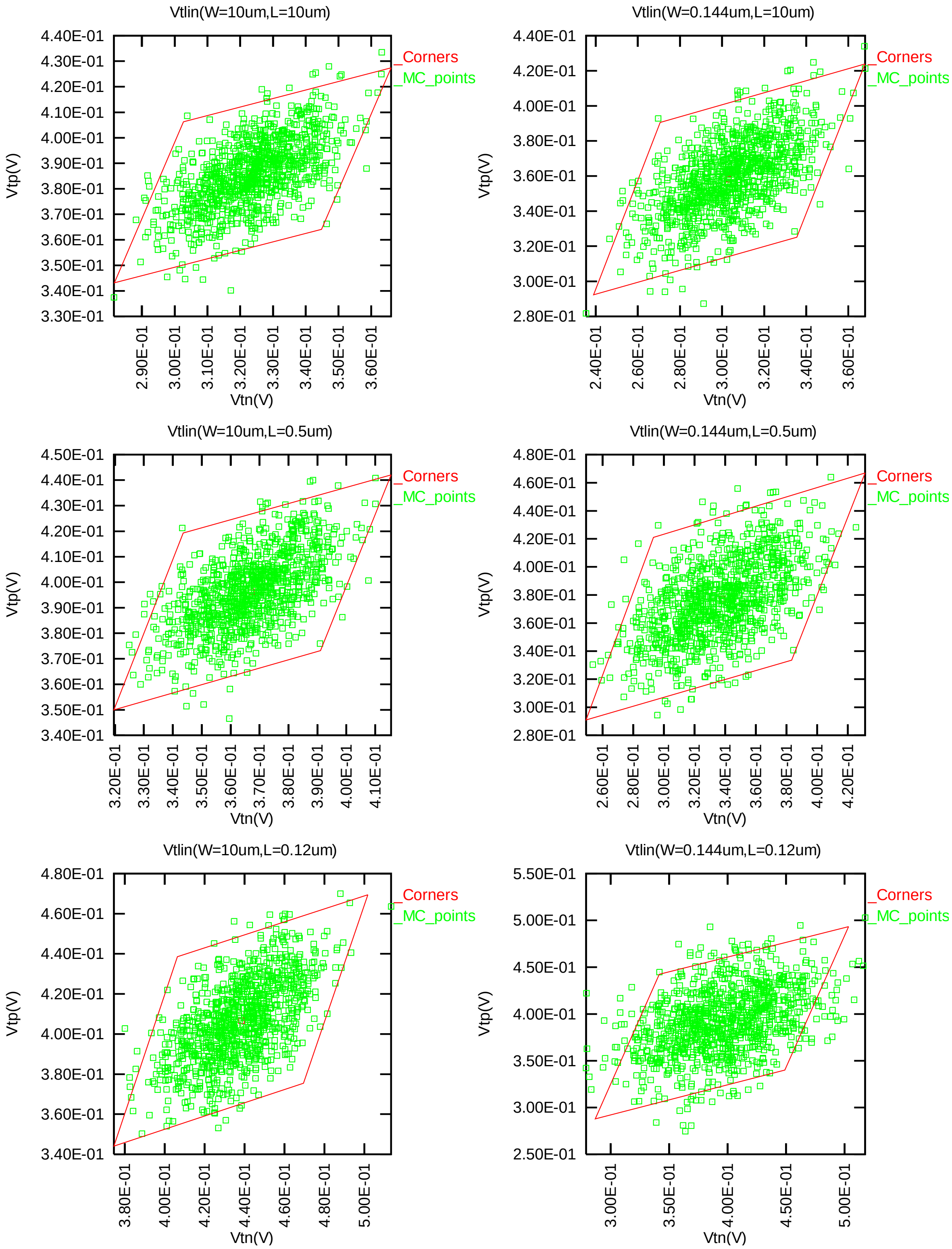


**Figure 2-1 Monte Carlo graph of threshold voltage at low drain bias**

## 2.2 Threshold voltage in saturation region VTSAT

The following figures show the Monte Carlo simulation result of saturation threshold voltage extracted at $I_D$ = 300nA * $W_{DES}$ * 0.9 / ($L_{DES}$ *0.9 + 0.012 μm) for NMOS and $I_D$ = -70nA * $W_{DES}$ * 0.9 / ($L_{DES}$ * 0.9 + 0.012 μm) for PMOS at 25 ℃. The dimension is after 90% shrink and 12 nm sizing channel length back.

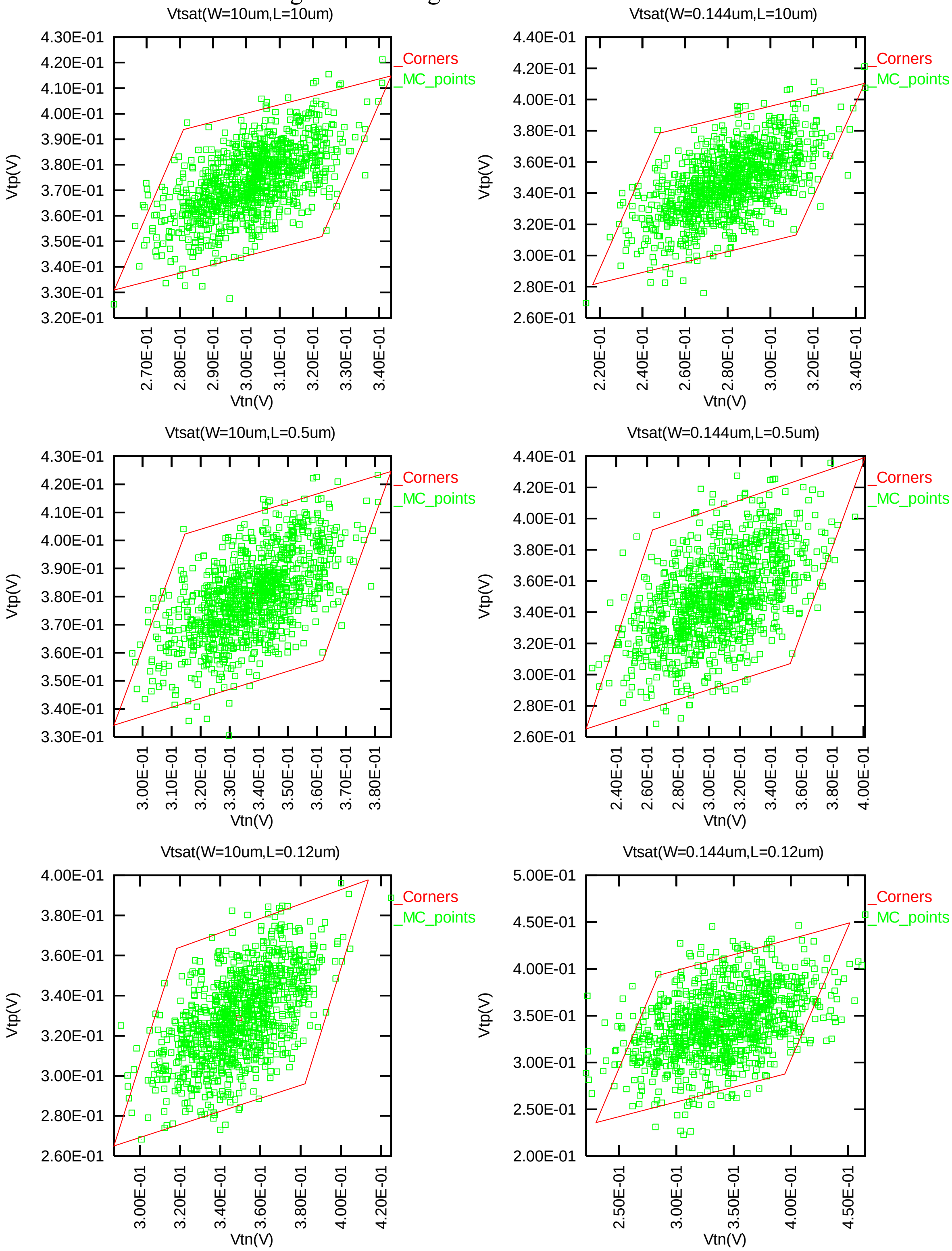


**Figure 2-2 Monte Carlo graph of threshold voltage at high drain bias**

## 2.3 Saturation Current IDSAT

The following figures show the Monte Carlo simulation result of saturation current at 25 ℃.

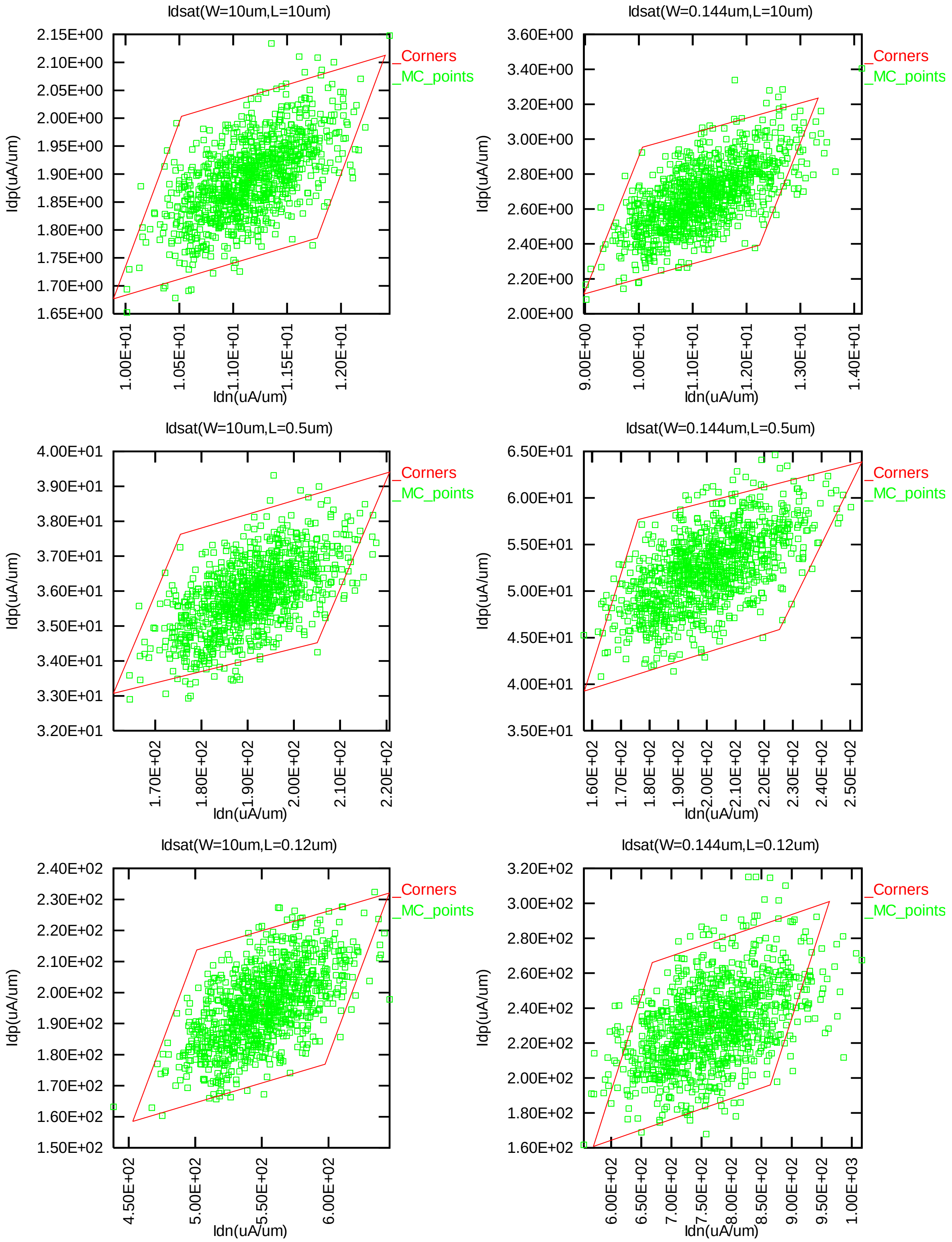


**Figure 2-3 Monte Carlo graph of saturation current**

# 3 Monte Carlo Dynamic Simulation Result for Process Variation

## 3.1 Ring Oscillator Circuit

To assess the dynamic performance of these devices, a ring oscillator circuit with the following configurations is adopted.

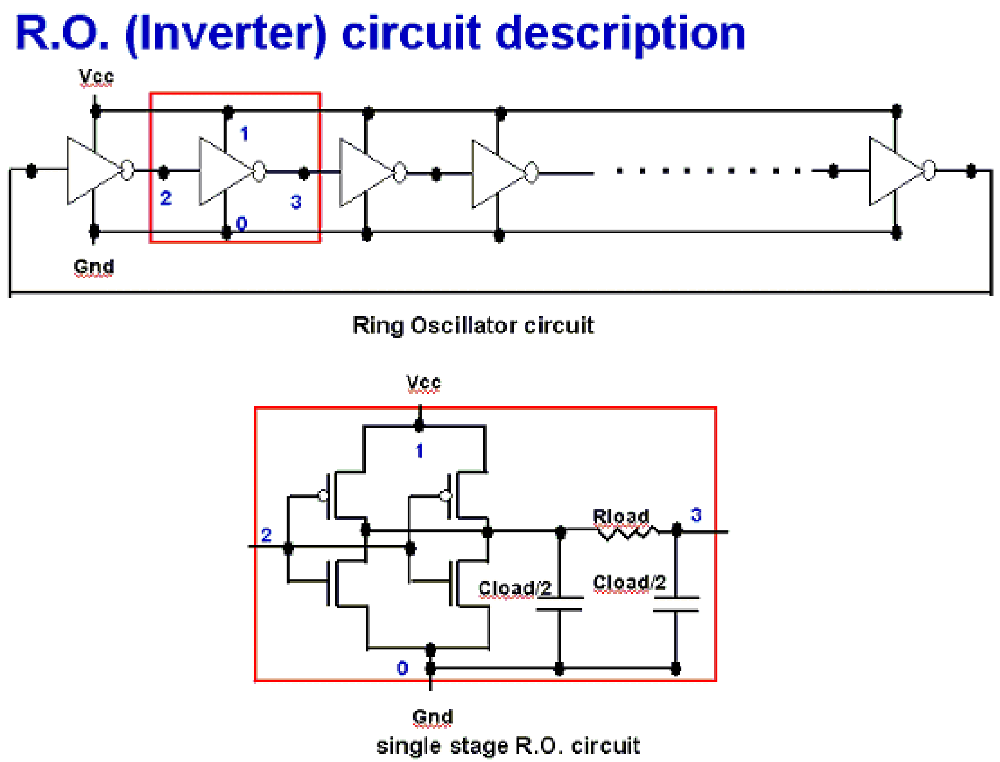


**Figure 3-1 Ring Oscillator Netlist description**

**Table 3-1 Ring Oscillator (Inverter) structure description**

| | |
|---|---|
| **R.O. (Inverter) Structure description** | |
| Contact size | 0.16 μm × 0.16 μm |
| Polysilicon gate to contact spacing | 0.12 μm |
| Contact to diffusion edge spacing | 0.06 μm |
| Lumped parasitic wiring resistance | 0 Ohm |
| Lumped parasitic wiring capacitance | 0 F/stage |
| **Inverter** | |
| Logic gate type | Inverter |
| Channel length | 0.12 μm |
| Channel width for n-ch MOSFET | 2.5 μm devices × 2 |
| Channel width for p-ch MOSFET | 3.75 μm devices × 2 |

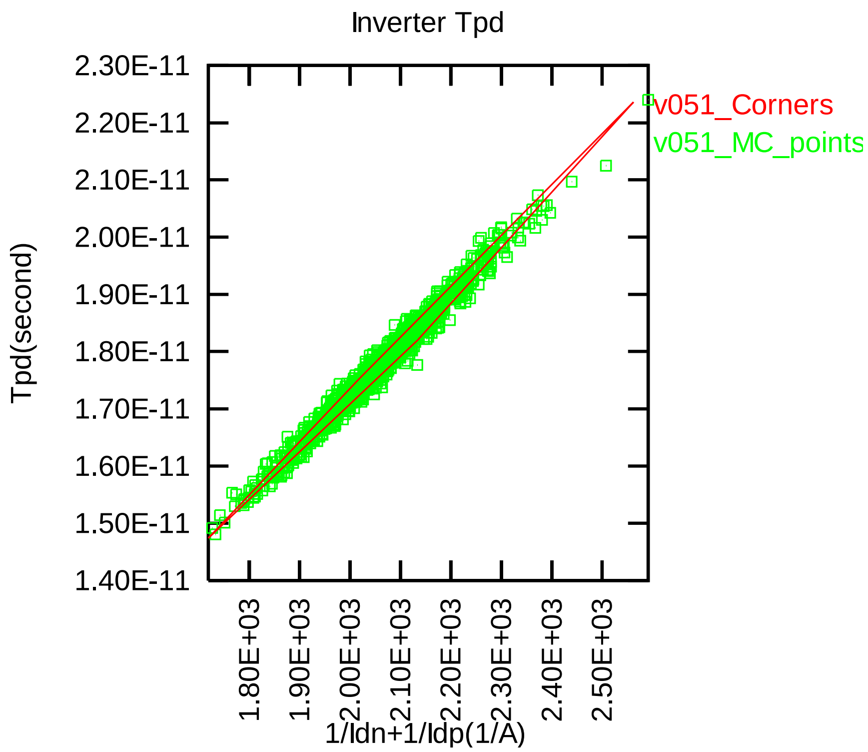


**Figure 3-2 Corner graph of ring oscillator**

# 4 Monte Carlo DC Simulation Result for Mismatch

## 4.1 Threshold voltage in linear region VTLIN

The following figures show the Monte Carlo mis-matching simulation result of VTLIN threshold voltage extracted at 25 ℃.

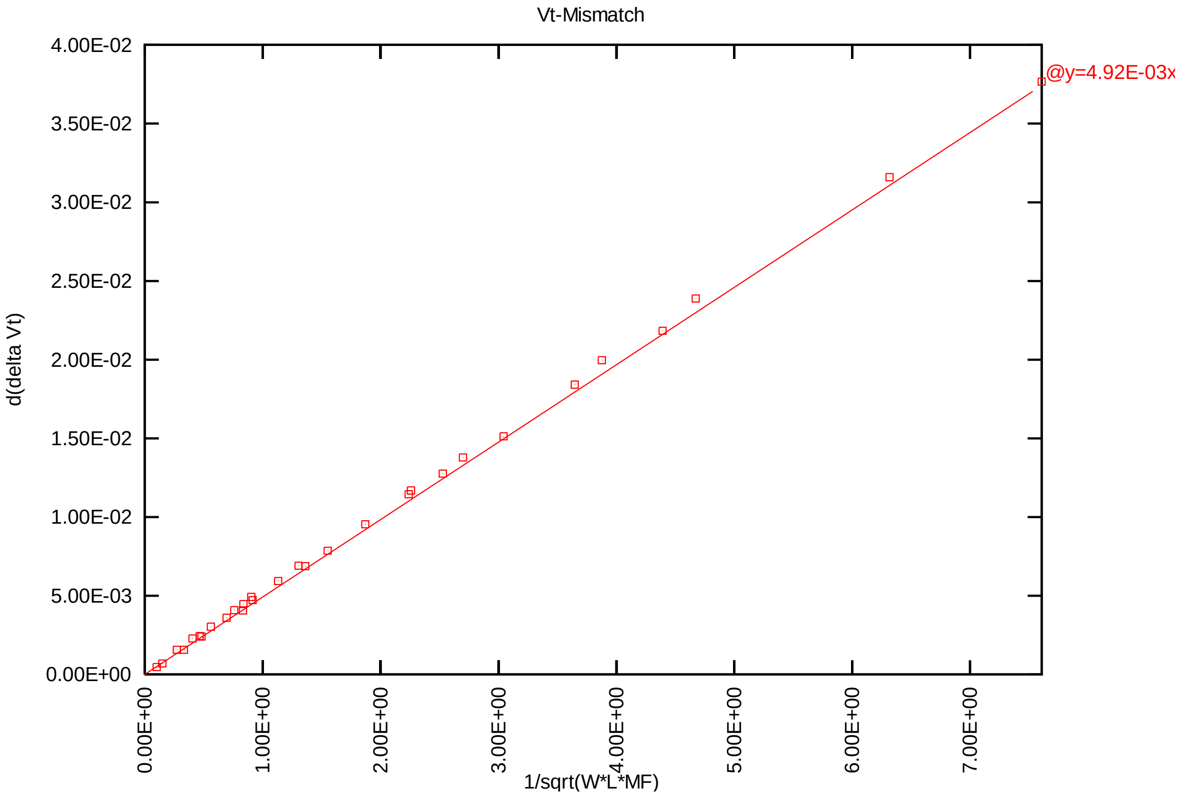


**Figure 4-1 VTLIN mis-matching simulation of 1.2V n-ch MOSFET**

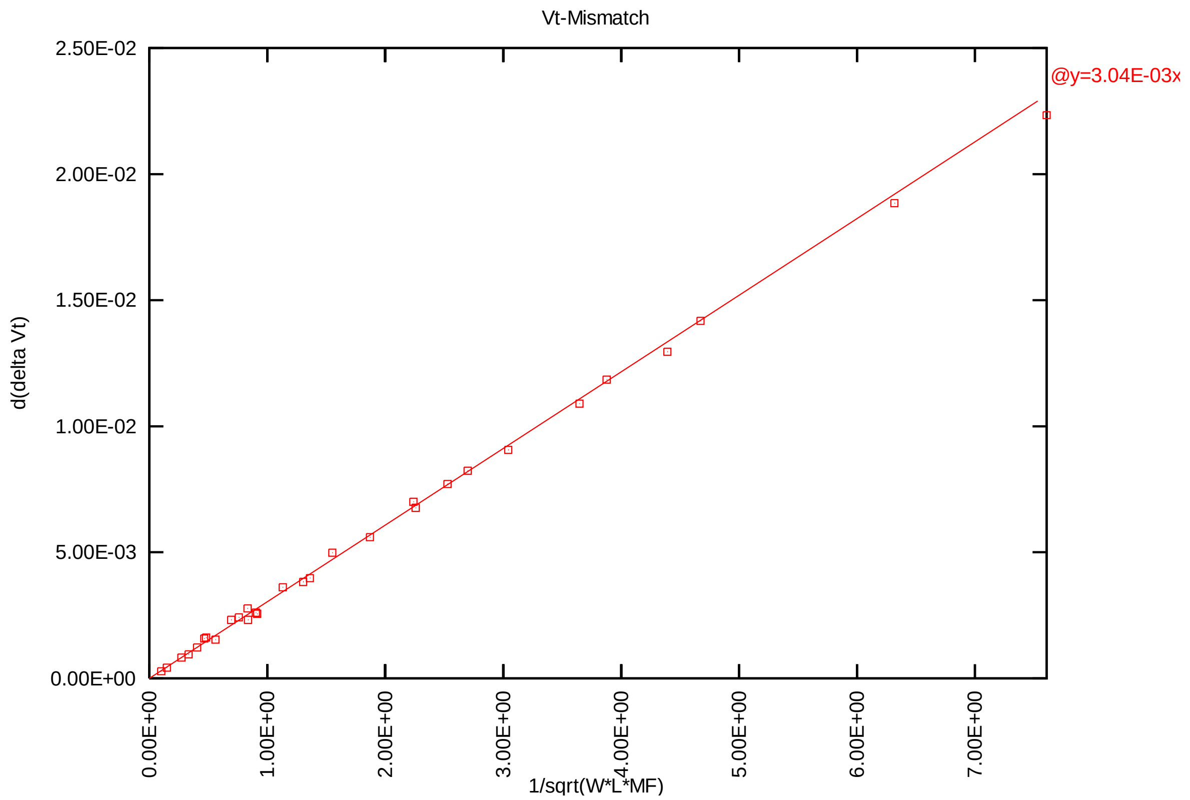


**Figure 4-2 VTLIN mis-matching simulation of 1.2V p-ch MOSFET**

## 4.2 Saturation Current IDSAT

The following figures show the Monte Carlo mis-matching simulation result of saturation current at 25 ℃.

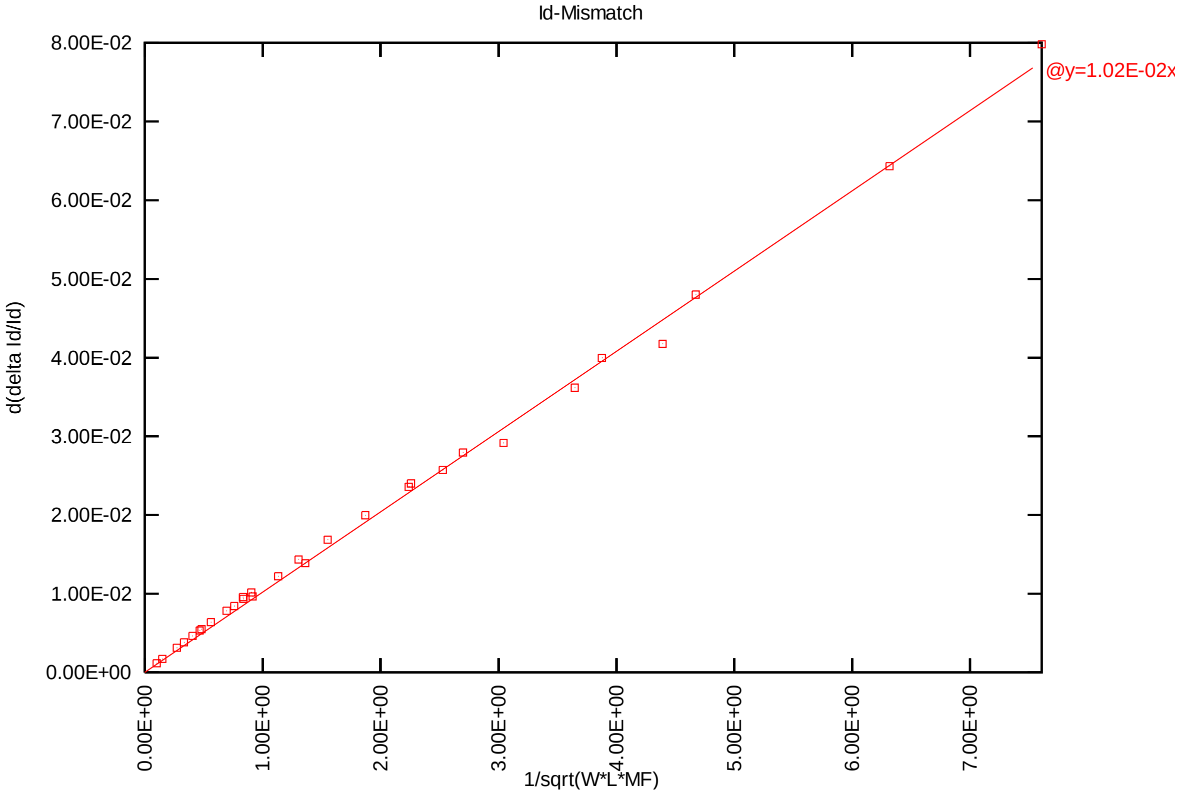


**Figure 4-3 Idsat mis-matching simulation of 1.2V n-ch MOSFET**

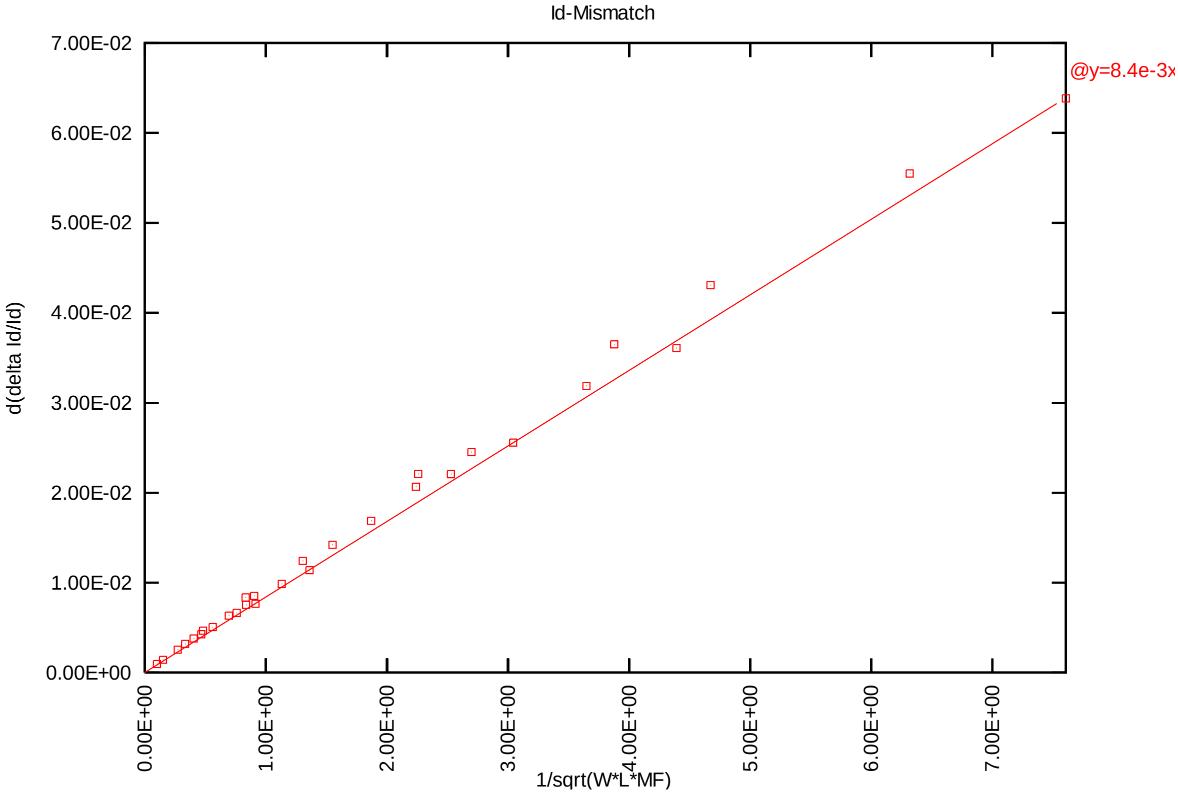


**Figure 4-4 Idsat mis-matching simulation of 1.2V p-ch MOSFET**

# 5 Appendix

## 5.1 HSPICE warning message

**Table 5-1 Hspice warning message**

| Item | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |

## 5.2 ELDO warning message

**Table 5-2 Eldo warning message**

| Item | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |

## 5.3 SPECTRE warning message

**Table 5-3 Spectre warning message**

| Item | Warning Message | Explanation |
|---|---|---|
| 1. | None | None |